



YISHUN INNOVA JUNIOR COLLEGE
JC 1 PROMOTIONAL EXAMINATION
Higher 2

CANDIDATE
NAME

CG

INDEX NO

COMPUTING

Paper 2 Practical

9569/02

26 Sep 2024

1 hour 50 minutes

Additional Materials:

Removable storage device with the following files:

- `PROMO_template.ipynb`
- `SCORES.TXT`
- `Database.db`

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.

All tasks must be done in the computer laboratory. You are not allowed to bring in or take out any pieces of work or materials on paper or electronic media or in any other form.

Approved calculators are allowed.

Save each task as it is completed.

The use of built-in functions, where appropriate, is allowed for this paper unless stated otherwise.

The number of marks is given in brackets [] at the end of each task.

The total number of marks for this paper is **60**.

Instruction to candidates:

Your program code and output for all the questions should be saved in a single file and downloaded as

PROMO_<your class>_<your name>.ipynb

For each of the sub-tasks, add a comment statement, at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

#Task 1.1
Program Code

Output:

In [2]:

#Task 1.2
Program Code

Output:

In [3]:

#Task 1.3
Program Code

Output:

- 1 A student intends to deposit \$ x at the beginning of every month into a bank account where $10 \leq x \leq 100$. The bank will pay a compound interest of 2% per month on the last day of each month.

Task 1.1

Write program code to:

- prompt the student to enter the amount x he intends to deposit
- validate that the input value is between 10 and 100, both inclusive
- if the input value is not between 10 and 100, print an error message "Invalid Input" and prompt the student to input the value again
- repeat until a valid input is obtained. [2]

Task 1.2

Write program code for the function `calculate(n)` to calculate the amount of money in the bank account at the end of n months, correct to 2 decimal places. [2]

Task 1.3

Write program code for the function `months(x)` to calculate the least number of months needed for the student to save at least \$2000, given that the student deposit \$ x at the beginning of every month.

The function should return a tuple containing the least number of months required and the amount of money in the bank account at the end, correct to 2 decimal places. [4]

- 2** A text file, SCORES.TXT, contains the results of the students for their recent examination.

Each line contains the student's name, Civics Group (CG) and the scores for English, Mathematics and Science, separated by commas.

Task 2.1

Write program code to:

- read the information from the text file and return a list of tuples
- each tuple contains the student's data in the following format:

```
(Name, CG, English_score, Mathematics_score, Science_score)
```

[6]

Task 2.2

Write program code to display the average Mathematics score of the students.

[2]

- 3** A research team from a university is studying the temperature variations within a day at a specific location.

An example of the hourly temperature readings for a particular day is as follows:

```
temp_data = [('18-Sep', '1:00AM', '24'),  
             ('18-Sep', '2:00AM', '24'),  
             ('18-Sep', '3:00AM', '24'),  
             ...  
             ('18-Sep', '10:00PM', '28'),  
             ('18-Sep', '11:00PM', '28'),  
             ('18-Sep', '12:00AM', '26')]
```

Task 3.1

Write program code for the helper function `get_temp(tup)` to return the temperature value of a particular tuple `tup` as an integer.

[1]

Task 3.2

Write program code for the function `insertion_sort(lst)` to implement the insertion sort algorithm to arrange the hourly temperature readings in the list `lst` in ascending order.

Test your program with the list of hourly temperature readings provided in the template.

[5]

The median is the middle value in a sorted list of values if there is an odd number of values in the list. Otherwise, the median will be the average of the two middle values.

Task 3.3

Write program code for the function `median(lst)` to use a sorted list of the hourly temperature readings to return the median temperature value as an integer.

[4]

- 4 The sum of the digits in the number 3156 is

$$3 + 1 + 5 + 6 = 15$$

Task 4

Write a recursive program code for the function `sum_of_digits(n)` to return the sum of the digits in the positive number `n`.

[4]

- 5 The distance for the inner most lane in a running track is 400.00 m. The distance for the lane next to it is 7.67 m longer and every subsequent adjacent lane is also 7.67 m longer.

Task 5

Write a recursive program code for the function `distance(n)` to compute and return the distance of the `n`th lane.

Use your program code to calculate the distance of the 8th lane in a running track.

[4]

- 6** The mail server at YIJC uses a hash table of size 20 to store the email addresses of all the users.

For example, the email address for the user 'abc' is `abc@yijc.edu.sg` and the sum of the ASCII values of the letters in the username is calculated as follow:

$$\text{ord}('a') + \text{ord}('b') + \text{ord}('c') = 97 + 98 + 99 = 294$$

Therefore, the hash value

$$\text{hash}('abc@yijc.edu.sg') = 294 \% 20 = 14$$

Task 6.1

Write program code for the function `hash(email)` to compute and return the hash value of the username. [3]

The hash value computed for the username is used as the index position to store the email address in the hash table. The following list of email addresses are to be stored in a hash table:

```
email_list = ['mrs_ho@yijc.edu.sg' , 'mdm_goh@yijc.edu.sg',  
              'mr_lee@yijc.edu.sg', 'mr_yeo@yijc.edu.sg']
```

Task 6.2

Write program code for the function `hashtable(lst)` to:

- initialise an empty hash table of size 20
- store all the emails in the list `lst` into the hash table according to the index positions found using the hash values of the usernames
- resolve any collision using the linear probing strategy
- return the hash table

Test your program code with the given list `email_list`.

[6]

7 The following is the first paragraph of a fantasy story:

```
paragraph = "In a land far away, there was a legendary kingdom  
known as Aurum, where gold, silver, and diamonds  
shimmered in every corner. The palace walls gleamed  
with golden bricks, while the streets were paved  
with silver stones that sparkled under the sun. The  
king, wore a crown decorated with diamonds. His  
throne, carved from pure gold, was said to radiate  
power and fortune to all who sat upon it."
```

A treasure hunter is searching for gold, silver and diamond in this fantasy land. He uses a dictionary data structure to store all the treasures he found.

Task 7.1

Write program code for the function `search(para)` to:

- search through the passage `para` looking for the word "gold", "silver" and "diamonds"
- count the number of times each of these words are found in the passage and store them in a dictionary
- return the dictionary.

Test your program code with the passage `paragraph`. Do not miss "gold" in the word "golden". [4]

After stealing all the treasures found, the hunter casts a spell to convert them into wood.

Task 7.2

Write program code for the function `cast_spell(para)` to:

- change all the word "gold", "silver" and "diamonds" found in the passage `para` to "wood"
- return the modified passage. [3]

8 The database `database.db` contains a table `books` with the following fields:

`Id, Title, Author, Publisher, Year, UnitPrice and Quantity`

Task 8.1

Write SQL Query command to display the `Title` and `Publisher` for all the books. [2]

Task 8.2

Write SQL Query command to display the `Title` and `Year` for all the books published after the year 2000. [2]

Task 8.3

Write SQL Query command to display all the publishers in the alphabetical order. [2]

Task 8.4

Write SQL Query command to display the total cost of all the books. [2]

Task 8.5

A new book has been purchased and the following information needs to be added into the database:

- Title: The Hidden Language of Computer Hardware and Software
- Author: Charles Petzold
- Publisher: Microsoft Press
- Year of Publication: 2000
- Unit Price: \$34.99
- Quantity: 2

Write SQL command to insert a new record into the table. [2]

END OF PAPER